AWS VPC

Virtual Private Cloud

VPC = Your own private network inside AWS cloud.

A **VPC (Virtual Private Cloud)** is a **logically isolated virtual network in AWS where you can launch AWS resources such as EC2 instances, databases, and load balancers**.

Main Components of VPC

1. CIDR Block (IP Range)
2. Subnets
3. Internet Gateway (IGW)
4. Route Table
5. NAT Gateway
6. Security Groups
7. Network ACL (NACL)

CIDR

CIDR (Classless Inter-Domain Routing) is a notation used to **represent a range of IP addresses using an IP address and a prefix length.**

CIDR format:

IP Address / Prefix Length

Example:

192.168.1.0/24

Private CIDR Ranges

Private networks usually use these CIDR ranges.

Range	CIDR
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

Subnet

A **Subnet (Subnetwork)** is a **smaller network inside a VPC.** It helps organize and control resources like **EC2, RDS, Load Balancers** inside a VPC.

Types of Subnets

1. **Public Subnet**

   A **public subnet** can communicate with the **internet**.

1. **Private Subnet**

   A **private subnet cannot be accessed directly from the internet**.

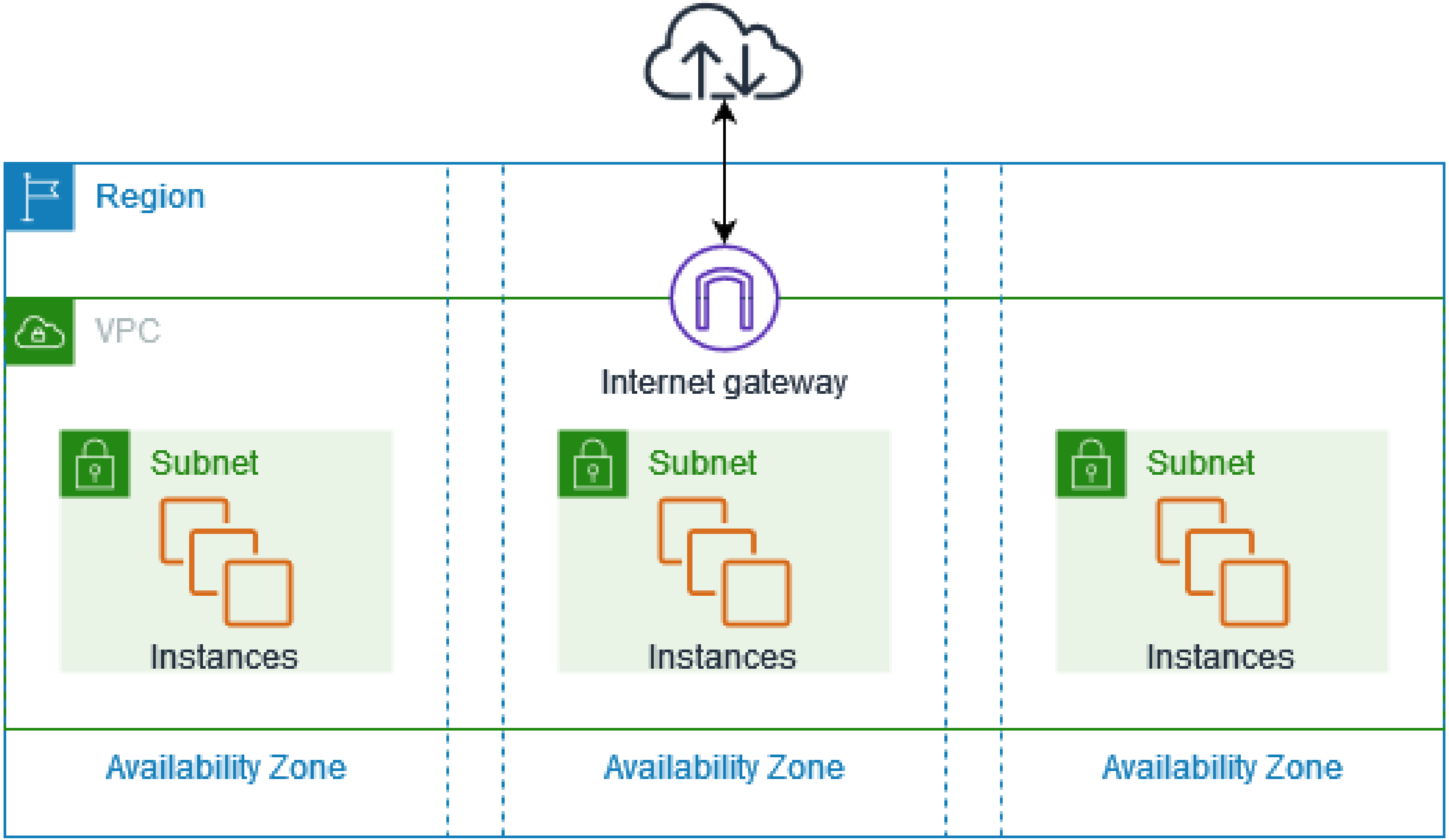
Internet Gateway (IGW)

An **Internet Gateway (IGW)** is a **VPC component that allows communication between your VPC and the Internet**.

Example:

EC2 → Subnet → Route Table → Internet Gateway → Internet

Without an Internet Gateway, your VPC cannot access the internet.


Region
VPC
Internet gateway
Subnet
Instances
Subnet
Instances
Subnet
Instances
Availability Zone
Availability Zone
Availability Zone

Route Table

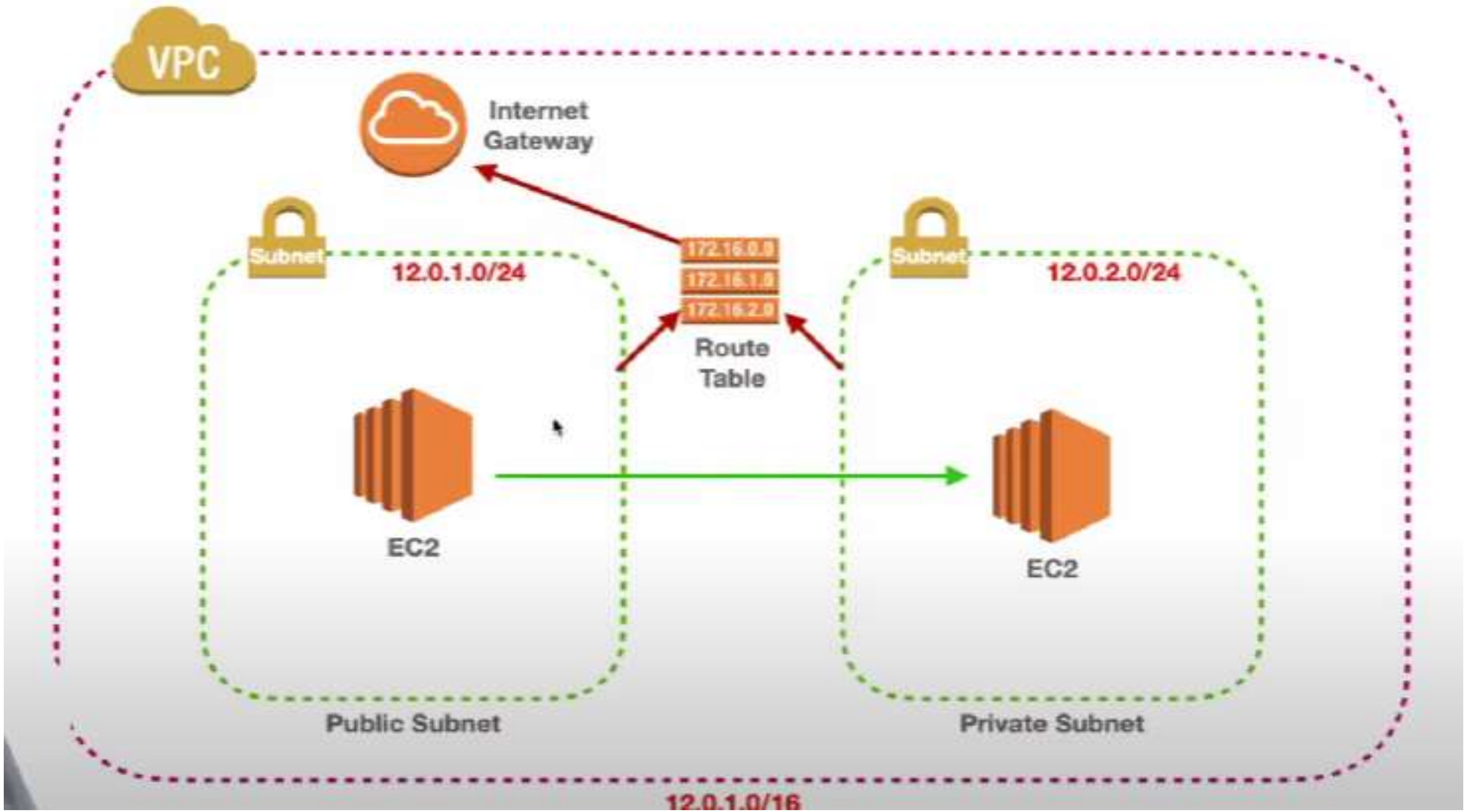
A **Route Table** is a set of **rules that determines where network traffic from your subnet or gateway is directed**.

Route Table Structure

A route table contains **routes**.

Each route has two parts:

- Destination : Ip address range
- Target : Where traffic should go

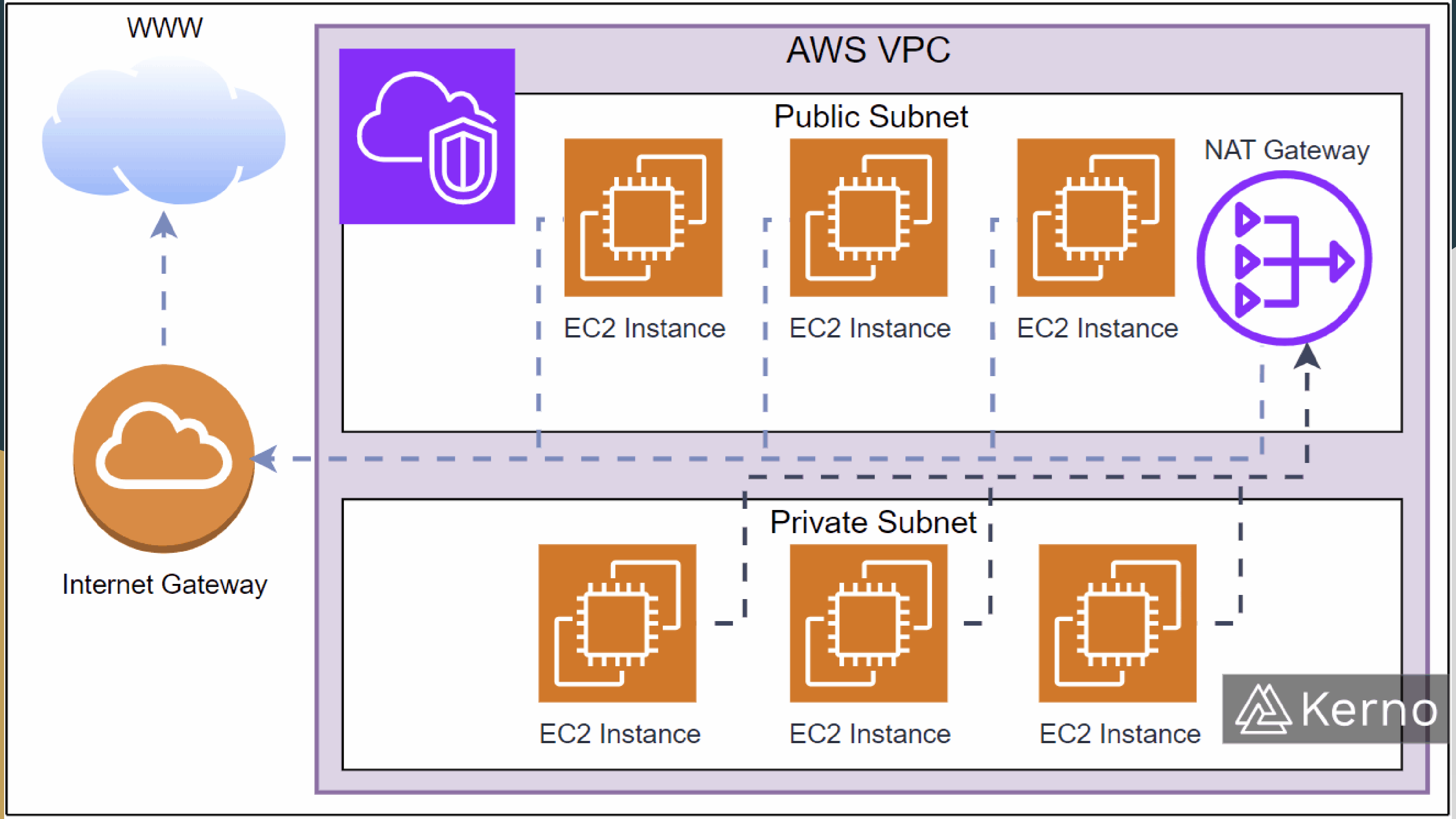

VPC
Internet Gateway
Subnet
12.0.1.0/24
172.16.0.0
172.16.1.0
172.16.2.0
Route Table
Subnet
12.0.2.0/24
EC2
EC2
Public Subnet
Private Subnet
12.0.1.0/16

NAT Gateway

A **NAT Gateway (Network Address Translation Gateway)** is a service that allows **instances in a private subnet to access the internet**, but **prevents the internet from directly accessing those instances**.

Use case:

- Download updates
- Access external APIs
- Install packages


WWW
AWS VPC
Public Subnet
EC2 Instance
EC2 Instance
EC2 Instance
NAT Gateway
Internet Gateway
Private Subnet
EC2 Instance
EC2 Instance
EC2 Instance
Kerno

Security Group

A **Security Group** is a **virtual firewall attached to an EC2 instance**.

It controls **incoming and outgoing traffic** to that instance.

Key Characteristics

- Works at **instance level**
- **Stateful**
- Only **Allow rules**
- Automatically allows response traffic

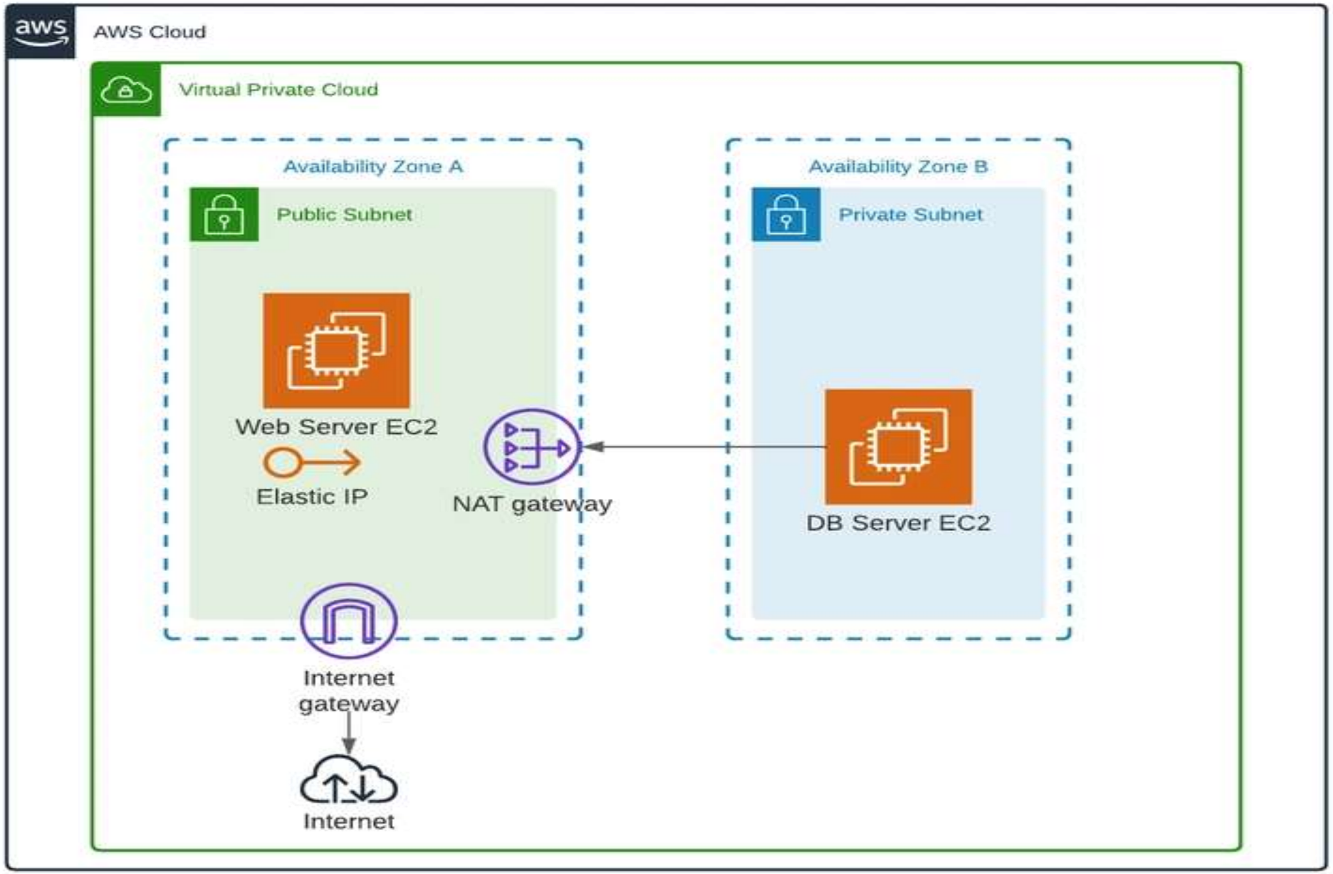
Network ACL (NACL)

A **Network ACL** is a **firewall attached to a subnet**.

It controls traffic entering and leaving the **entire subnet**.

Key Characteristics

- Works at **subnet level**
- **Stateless**
- Supports **Allow and Deny rules**
- Must configure **both inbound and outbound rules**


AWS Cloud
Virtual Private Cloud
Availability Zone A
Public Subnet
Web Server EC2
Elastic IP
NAT gateway
Internet gateway
Internet
Availability Zone B
Private Subnet
DB Server EC2